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Sahayak: Affordable AI enabled Assistive Technology for Intellectually Disabled

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***Abstract*— Children with Intellectual Disabilities often face challenges in traditional learning environments. This can lead to frustration and hinder their academic progress. They need specialized therapy or special educator sessions to bridge this gap. Advancements in technology, especially in Artificial Intelligence, offer exciting possibilities to bridge this gap. This paper demonstrates this possibility by the development of a mobile assistive technology app, 'Sahayak', that utilizes Augmented Reality (AR) and Artificial Intelligence (AI) to create an engaging and personalized learning experience for children with Intellectual Disabilities especially in the area of reading and comprehension. The app leverages AR to overlay interactive elements onto the real world through a smartphone or tablet camera. This allows children to interact with educational content in a fun and immersive way. AI plays a crucial role by tailoring the app's functionalities to each child's specific needs. The app can assess a child's strengths and weaknesses, and then curate learning modules using Machine Learning and Augmented Reality that target specific areas for improvement. This personalized approach empowers children to learn at their own pace and fosters a sense of accomplishment.**

Keywords—Augmented Reality, Intellectual Disability

I. INTRODUCTION

Children with Intellectual Disabilities (ID) face a variety of challenges in the classroom, such as difficulty with reading, writing, math, or organization. These challenges can lead to frustration, decreased confidence, and a negative impact on academic achievement. The ubiquity of mobile devices has led to the development of a vast array of mobile applications that help in eLearning. However, use of adaptable personalized apps that can act as assistive technology (AT) for children with Intellectual Disabilities needs much exploration. AT is defined as any device, piece of equipment, or system that helps bypass, work around, or compensate for an individual's specific learning deficits.

Reading and writing disabilities are common in most children with ID. They cannot naturally process written words. Sometimes they may not be able to understand the meaning of simple words. Due to issue with short-term memory, they cannot remember new words easily. Their reading and writing speed and accuracy may not be the same as other children of their age. In some cases, children may not be able to pronounce the words correctly, hence making it difficult for them to read. Such reasons make it gruelling for children with Intellectual Disabilities to strive in a classroom full of normal children, and hence, in the society. This could result in academic failure, emotional problems, and adjustment difficulties [5]. Today, the world is advocating for inclusivity, starting from school. When children are included in school, use of appropriate AT to enable their integration and adaptation is very crucial. This need is so understood by the operating system designers that there are inbuilt accessibility tools put specially for differently abled. The most common tools are captioning tool, immersive readers, talkback etc. This paper explores the use of Artificial Intelligence and Augmented Reality to enrich the learning experience of the students.

Augmented Reality is a technology where virtual components are superimposed on the real world to create an altered reality, to help users to interact with an environment supplemented with virtual information. The ability to overlay digital information onto the real world in real-time opens up endless possibilities for enhancing our lives and improving productivity and safety in various fields. Application of Artificial Intelligence has penetrated almost all areas of science and emergence of generative AI has given a new dimension to it. The use of Large Language Models for chatbots have been seen in the market and there is great progress in video and image content creation via Artificial Intelligence. AT developed using these technologies can provide the user with a richer user experience.

This paper explores the creation of prototype mobile app 'Sahayak' using technologies like Augmented Reality (AR) and Artificial Intelligence (AI) to help children with Intellectual Disability. The tool Sahayak is built to help children with reading and comprehension difficulties while reading printed text. The application scans text using camera and helps the user read it in the speed he or she requires by highlighting current word being read. It uses AR to help in comprehension by supplementing information like synonyms, definition, parts of speech etc. about the text, for each and every word. Instead of using a static dictionary, the app explores the use of AI-generated images for a word to enhance visual learning. The app also explores the use of more tactile learning experience by using 3D models. The 3D models are displayed on the user's device and as augmented reality using Google's ARCore for Android.

The paper is organized as follows. The motivation to work in the area of AR assistive technology is explained in section II. The review of existing applications/research and gaps identified are listed in Section III. The technical details of the proposed Android application have been discussed in detail in

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design Section IV. Section V gives a brief discussion of the challenges and Section VI concludes the paper.

II. Motivation

In recent years, our classrooms have evolved to be inclusive classrooms where differently abled children are welcomed along with neurotypical children. The adaptations in curriculum are minimal when teachers try to adapt their pedagogy. The limitations in the implementation can be practically handled through the introduction of AT tools to help them. AT tools can use multi-sensory approach in teaching children especially reading and writing. AR has great potential to enhance accessibility and independence without having technical complications of using Virtual Reality (VR). Now that mobile technology has become very affordable, mobile based tools for low-cost reading interventions are very much needed.

This paper explains our work in developing a proof-of-concept software that contains a visual dictionary for displaying 3D images of objects, to give a more real-life experience to the child, along with AR-based audio plus visual output. Through this tool, we aim to provide an immersive multi-sensory learning experience to overcome the challenges faced in classroom by children with ID.

III. Related Works

In an inclusive society it is important to provide training and enabling technologies to help intellectually disabled people live as independently as possible. In a clinical setting this is done by a therapist catering to particular needs of individual. Period like COVID has emphasized the need to provide supplementary help using technology to continue these treatments. Here are different type of training and AT tools that are needed to empower the individuals with ID. The literature identifies need to develop tools for

1. Communication and Languages - AAC (Augmentative and Alternative Communication) apps to help in communication, text to speech app to help in reading.
2. Daily Living Assistance - Reminder apps, Easy prompting with smart home technologies
3. Educational and Learning - eLearning tools that cater specific to the learning issues
4. Health monitoring and Safety management
5. Recreation apps.

In this paper we specifically look at enriching the learning experience of individuals with ID. Reading and writing problems are commonly seen in children with Intellectual Disabilities. Computer-based applications are being used to change conventional environments to new interactive ways of learning. AR enables a multi-sensory approach to enhance the learning experience of these children. [10] highlights how AR technology has been used to enhance the learning experience of the students with LD, while shedding light on the potential of AR to provide inclusive and effective learning environments for this student population, thereby contributing to a broader discussion of technical and higher education. Several applications have been developed to aid the education of children with disabilities. This section gives a comprehensive review of the related works in this area.

Assistive technology such as text-to-speech software is available as part of accessibility tool kit in all operating systems. This is mostly useful for children with visual disabilities. Augmenting information to the text visually may help individuals with ID to improve their reading skills. As majority of children with ID are visual learners, image associated with text can make learning process easy. A sample set of AT tools used in market that are based on AR were reviewed, with their features, are mentioned in Table I.

Sr. No.	Source	Features
1	Kurzweil 3000 [9]	Text-to-Speech, Speech-to-Text, Grammar correction, OCR, AI based solutions
2	Avaz Inc [2]	Text/Image-to-Speech, Augmentative and Alternative Communication
3	AR Story Book [4]	A book containing printed text having triggers to show additional 2D or 3D layers of content using AR
4	ASD Simulator [3]	AR based simulator to replicate ASD people's vision using a computer based visual simulation environment
5	AR for mastering vocabulary [1]	Scanning QR code to display 3D image of associated object in the environment
6	AR Ole Cierraojos [8]	Learning based AR game that promotes reading comprehension by projecting a 3D game based on the written text on the scannable page.
7	AR assistive technology for handicapped people [7]	Embeds pre-recorded audio with image of AR tag when an AR tag is scanned
8	disruptED	An IOS app that allows learners to select either augmented or virtual reality to engage with the books, which include visual and auditory prompts in the directions.
9	Autism Glass [11]	A wearable device that uses AI technology to provide real-time social cues and feedback to children with autism. The glasses use computer vision to recognize emotions and provide social prompts to help the child navigate social situations.
10	Brain Power [12]	An app that uses augmented reality and AI technology to help children with autism learn and develop social skills. The app features interactive games and activities that help children practice social interactions and communication skills.

TABLE I. AR-Based Tools for reading

These are the most extensively used AR apps by special educators, specially abled children, and their parents. There are a limited number of immersive technology tools for children

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with ID. Most of them are intended for entertainment and recreational activities, and ones specifically geared towards early elementary education continue to be limited. We can see that most of them use only traditional AR technologies or none at all, suggesting that the use of AR in the field of assistive technology is scarce. The ones that do make use of Augmented Reality have a very limited scope and still use conventional AR technologies. Another way of integrating Augmented Reality into children's assistive learning is using VR Headsets. Firstly, these headsets are extremely expensive, and hardly affordable. Secondly, an alarming concern with the use of headsets is that of safety. The size of the headsets does not fit well on little heads and often opens the door for dangerous falls or collisions. Some students report feeling sick or nauseous when using any VR headset. It should also be noted that most of such AR based Assistive Technology apps in the market are all paid, making this technology inaccessible to children who would need it. Majority of the tools discussed in this section use traditional AR technologies and with little emphasis on teaching elementary education to children with ID. Hence, looking at the current AR-based assistive technologies, it suggests a need for more accessible, high-tech, free-of-cost product that benefits the children with Intellectual Disabilities.

IV. Tools

A. Requirements

As a preliminary work it was decided to explore development of a tool to improve the reading and comprehension of students with ID by using AR and AI. The input the system is the photo of the text that the user takes, and output is text enriched as per required by the user.
The requirements for the tool are listed below.

1. Scanner-The individual should be able to read printed text that is in a screen or book with the assistance of tool. The individual should be able to scan the text using camera attached to the mobile. Currently the text is limited to printed text.
2. Reading Tracker- Software that can isolate words and help the individual to trace the word and read at his/her pace.
3. Text Enricher – The individual should be able to understand the word better in two contexts.
    a. Grammatically – The part of speech represented by word.
    b. Semantically – Meaning of word in audio and visual (2D and 3D) cues.
4. Read Aloud - The app should provide read aloud facility for the text to allow children to listen to the words.
5. Usability – The proposed application should be easy for the users to carry it along so that they can easily point to any text they need help with.
6. User Interface – The user interface should be very intuitive.

B. Design

The usability requirement points towards a mobile app-based design. Use of mobiles as AT is very accepted and it is a low-cost option as it reuses existing device. The proposed design of the tool is given in Figure 1.

The tool consists of four modules - Text Recognizer, Sentence Processor, Text Enrichment Module and Reading Tracker. Text Recognizer Module captures text to be processed through camera and the module converts the scanned text into words. This is done using Google Play Service's Vision engine. Reading Tracker Module provides an interface to display the captured information in the way user requires. It is implemented using the TextToSpeech engine by Android. This module acts as GUI for the app. Sentence Processor takes the output of the Reading Tracker and processes the sentence to identify the Parts of Speech, Synonyms and Meanings of the words in the sentence with the use of Collegiate and Thesaurus API. The option to view synonyms of the word will help users to improve their vocabulary and comprehension skills, and the identification of the part of speech for each word will help them to understand the grammatical structure of sentences. The Text Enrichment Module consists of two parts: AI module and AR Module. AI module creates static images for words dynamically fitting to the context of the sentence by generating images for a word as a prompt using an ML based image generator created using Stable Diffusion Pipeline. The AR module retrieves AR models using Google's ARCore and SceneViewer that most appropriately matches the word to enrich user experience by viewing 3D models in their space. This feature is particularly useful for users who learn best through visual and tactile experiences, as it provides a hands-on learning experience that enhances their understanding of the world around them.

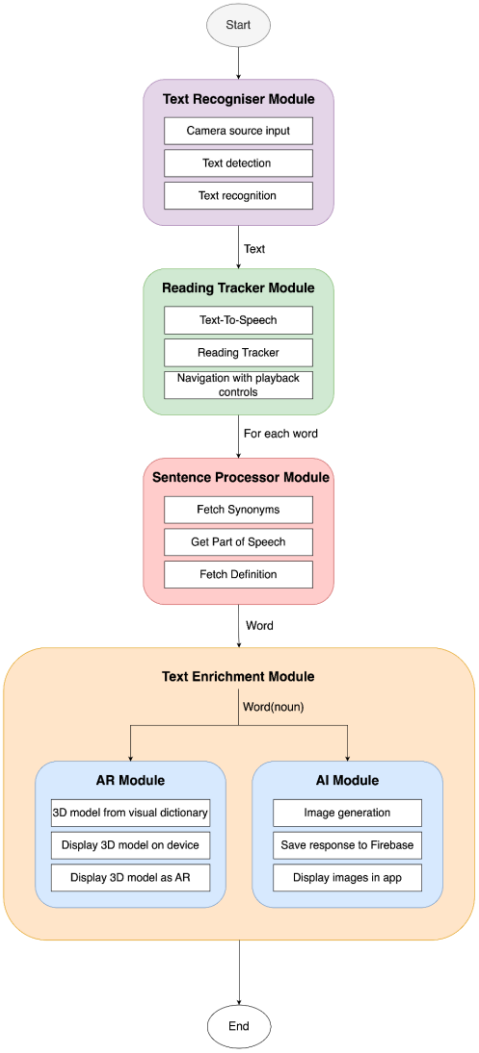


Figure 1. High level Design

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C. Proof of Concept

The proof-of-concept application “Sahayak” is built on Android Studio's IDE using Java for all background processes and as a bridge between back-end and web-APIs. The text detection and recognition were done using Google Play Service's Vision engine. The read-aloud feature is implemented using the TextToSpeech engine by Android. APIs have been used to fetch information regarding the text. An API for an image generation AI model has been created and hosted using Python and FastAPI framework. The 3D models are rendered using ARCore by Google.

The android application is developed on Android Studio using XML and Java for front-end and back-end, respectively. The app uses a simple yet engaging UI suitable for young children. Camera is used to capture the typed text live and the camera preview is displayed using the SurfaceView. The Vision engine provided by Google Play Services for Android is used to detect text from the camera source. The detected text is processed in TextRecognizer and displayed by the Reading Tracker in the app. Next, this processed text is taken to another page where it is read word by word at a slow rate using the TextToSpeech engine by Android. The reading can be played, paused, forwarded, rewound, and replayed by navigating through the words. For every word being read, the synonyms, definitions, and part of speech is displayed by fetching this information from the Collegiate and Thesaurus API by breaking down its output.

Next, we generate images for the word as a prompt using Machine Learning (ML) image generator created using Stable Diffusion Pipeline. This Python model uses a deep learning library called PyTorch to generate an image based on text input. The pre-trained model has been used to generate an image based on the text input by running it through a process called Stable Diffusion. This process essentially creates a gradual blending of pixels to generate an image. The PyTorch library has a class called Stable Diffusion Pipeline, specifically in the library's "Fairscale" module. This technology enables the creation of high-quality images from text prompts utilizing a method referred to as "Stable Diffusion". The process entails updating an image repeatedly over multiple time steps while incrementally introducing additional noise into it during each step. Gradually produced is a blurred depiction that can be refined and perfected for achieving exceptional clarity and finesse. Moreover, the Stable Diffusion Pipeline offers its own distinct mechanism useful for implementing the Stable Diffusion technique that can be used with pre-trained models. It takes a text prompt as input and generates an image based on that prompt. This class has several configurable options that can be used to customize the generated images, such as the level of detail, the amount of noise to add, and the size of the image. Any number of images can be generated using this model. The images are fetched by creating an API that gives image output for a given prompt.

These images provide further description of the word being read. These generated images are also saved in an online database (Firebase) for further use, if any. Application also contains a visual dictionary for displaying 3D images of objects, giving a more real-life experience to the child, along with AR-based audio plus visual output. Through this tool, we aim to overcome the problem of static dictionary which will fail when new words are seen. This tool intends to give a multisensory, immersive reading experience to the children with Intellectual Disabilities, in the areas of reading and writing.

We tried to extend the experience by including 3D models in AR. An online visual dictionary consisting of 3D models of basic words for all shapes, foods, instruments, animals, etc has been created. If the word being read is present in the visual dictionary, its 3D model is fetched and is displayed as augmented reality using Google's ARCore and SceneViewer. If the user's phone does not support AR, even then, the 3D model can be viewed on the mobile screen. This gives a more real-world and tactile learning experience.

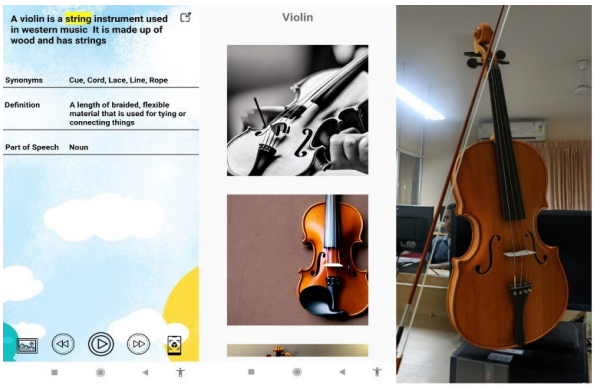
Currently we could not identify any AI model API that creates 3D models dynamically. So, this module has been tried in a more static fashion using a built-in library. If the 3D model is not available for a certain word, the application informs the user that the model is not currently available but will be added in the future. Additionally, the application stores the word and its demand in the back-end server. The demand for words without a 3D model can be tracked to identify which words are most in demand by users. Based on this data, the development team can prioritize the creation of 3D models for those words and update the visual dictionary accordingly.

The text from camera input is detected using the Vision engine provided by the Google Play Services for Android. The text recognition is done using the TextRecognition library in Android. The app uses The Collegiate and Thesaurus API to fetch synonyms, definitions, and parts of speech for a word. The Android TextToSpeech Engine is used in the implementation of reading tracker to read aloud the text.

A Python model API has been created using the FastAPI framework, which uses a deep learning library called PyTorch to generate an image based on text input using Artificial Intelligence. It uses a pre-trained model to generate an image based on the text input by running it through a process called Stable Diffusion. ARCore, a platform by Google to build Augmented Reality, has been used to implement AR in the app. The 3D models are rendered and displayed using SceneViewer. Sample Screenshots of reading tracker, AI generated image and AR image is given in Figure 2.

V. Discussion

The implementation of the project was challenging due to the high cost associated with most of the platforms and technologies used for AR development. The field of Augmented Reality is still relatively new, and the development of AR technologies is slow and stagnant, despite its vast potential uses. Free text recognition libraries like Vuforia, which are commonly used in Unity for AR development, have been deprecated. Moreover, many of the SDKs used with Unity for AR development are paid, making them inaccessible for some developers.



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Figure 2. Screenshots of reading tracker

We used Android Studio for development to overcome these challenges. Built in libraries of Android Studio and Google's AR development platform, ARCore was used in this project. The libraries and SDKs used for AR development in Android are available for free thus making the development process more accessible and cost-effective.

The technical specifications needed for a device to function with augmented reality (AR) include:

- A high-resolution camera to capture clear images.
- An accelerometer to provide data on the user's position and direction.
- The ability to connect to a high-speed internet connection.
- A well-lit environment for optimal use.
- A flat surface for displaying 3D models in AR.
- Stable placement for accurate text scanning.

Most of these requirements are commonly available with today's mobile technology advances both in terms of device as well as connectivity.

In the development of app, we used the stable diffusion pipeline for the image generator. We used individual words to generate image, but words can have different meaning in different context. Some of the images generated were wrong. A better approach will be to generate image based on complex prompts. Future work can explore how to generate more appropriate prompts to ensure that image is generated as per the context of reading.

As of the knowledge of authors there are only limited AI models developed to autogenerate quality 3D models from image. The models need extra information than just image. Learning experience can be improved if the image generated and text can be used together to improve generation of 3D models. Exploring the object in 3D will provide a semi tactile experience for the learning improving multi-sensory perception.

The individuals with ID are a minority when compared with neurotypical individuals. As the market for AT tools is limited, and the affordability for buying such tools are also limited, there is a need of development in open-source area. As in the case of Sahayak, developers need to focus on building affordable tools using existing technologies and equipment that individuals use.

VI. CONCLUSION

This paper demonstrated the feasibility of building a mobile based tool with AR and AI as an Assistive technology tool for helping individuals who struggle with reading and comprehension. The tool is provided with visual cues to improve comprehension of reading content. The prototype of the proposed model "Sahayak" was built to show the feasibility of the proposed design.

Augmented information in tools like Sahayak can make learning more interesting for children. As the tool is designed as mobile app it is easy to carry, and student can use it to supplement information missing in traditional classroom while reading.

As future work, Sahayak can be improved further to provide multiple language support using Google Translate APIs. The visual dictionary is currently hard coded for 3D models. This needs to be enhanced to include more complex concepts and objects. We intend to publish the application on Play Store and receive real time feedback from users.

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